Assignment 5 Solutions

1. val bar : 'a foo → 'a foo → 'a
   val baz : 'a foo → 'a
   val out : int

   So todo : int foo. There's pretty much only on choice,

   ```
   let todo = Foo (fun _ → 42)
   ```

   Then

   ```
   baz todo =
   baz (Foo (fun _ → 42)) =
   bar (Foo (fun _ → 42)) (Foo (fun _ → 42)) =
   (fun _ → 42) (Foo (fun _ → 42)) =
   42
   ```

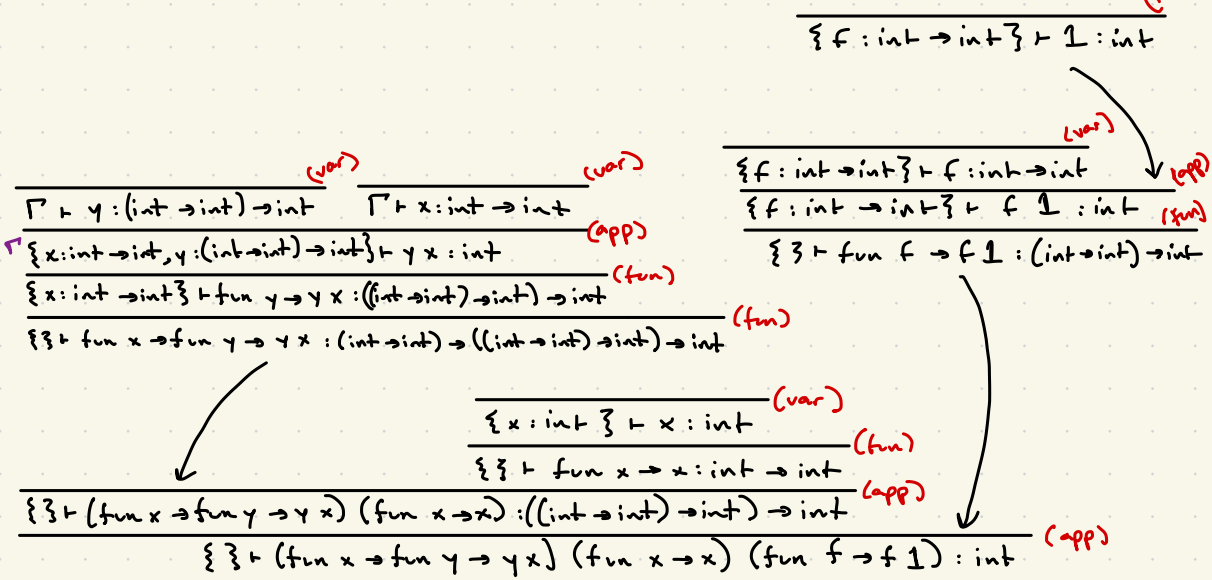
2. ```
   type task =
     | Apply of (int → int → bool list) * int
     | Flip of bool list
     | Filter of { pred : int → bool → bool; elems : bool list }
     | Reverse of bool list
   ```

3. val foo : `bool list → int list`

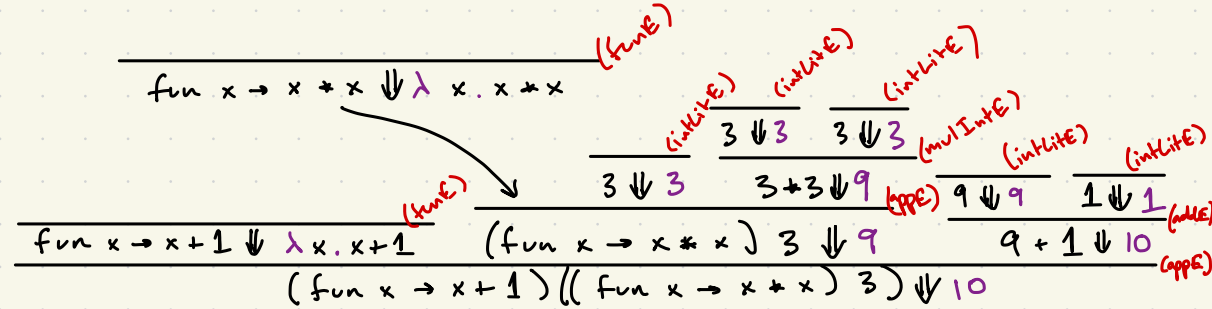
   foo produces a list of integers the same length as the input list. Starting at 0, true increments the next value and false decrements it (bar does the same).

   So the output is `[1; 2; 3; 2; 1; 0; -1; 0; 1]`.
   ```

4.


(intLit)
{f : int → int} ⊢ 1 : int
(var)
{f : int → int} ⊢ f : int → int
(app)
{f : int → int} ⊢ f 1 : int
(fun)
{} ⊢ fun f → f 1 : (int → int) → int
(var)
Γ ⊢ y : (int → int) → int
(var)
Γ ⊢ x : int → int
(app)
{x : int → int, y : (int → int) → int} ⊢ y x : int
(fun)
{x : int → int} ⊢ fun y → y x : ((int → int) → int) → int
(fun)
{} ⊢ fun x → fun y → y x : (int → int) → ((int → int) → int) → int
(var)
{x : int} ⊢ x : int
(fun)
{} ⊢ fun x → x : int → int
(app)
{} ⊢ (fun x → fun y → y x) (fun x → x) : ((int → int) → int) → int
(app)
{} ⊢ (fun x → fun y → y x) (fun x → x) (fun f → f 1) : int

5.


(funE)
fun x → x * x ⇓ λ x . x * x
(intLitE)
3 ⇓ 3
(intLitE)
3 ⇓ 3
(intLitE)
3 ⇓ 3
(mulIntE)
3 * 3 ⇓ 9
(intLitE)
9 ⇓ 9
(intLitE)
1 ⇓ 1
(addE)
9 + 1 ⇓ 10
(funE)
fun x → x + 1 ⇓ λ x . x + 1
(appE)
(fun x → x * x) 3 ⇓ 9
(appE)
(fun x → x + 1) ((fun x → x * x) 3) ⇓ 10